

Team Number: 44

Project Title: Breast Cancer Stage Classification with miRNA

Report Date: 9/01/25

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES or NO; if NO, explain.

Yes

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.

This week, we did not hold a formal team meeting. We coordinated daily through our team's Discord server to discuss individual and overall progress, assign tasks, and resolve questions or issues.

2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.

Yes. The entire team (Joshua Adegboyo, Anika Baro Villa, Nicholas Campos, Brehon Parker) met with Dr. Ali Ibrahim on Monday, September 1st, 2025, at 9 p.m.

3. Describe verbally the tasks completed the past week and the challenges faced.

This week (Week 3) our team's main goals were to clean the miRNA expression files and to link them with the clinical metadata so the dataset could be used for feature selection and modeling. Dr. Ali Ibrahim decided that we should all complete these steps together so that everyone would become familiar with the dataset, the structure of the raw files, and the MATLAB scripts he developed from his previous research.

We focused on cleaning and organizing the miRNA expression files from TCGA and then linking them with the corresponding clinical metadata. Using MATLAB, we extracted the normalized expression values, removed files that were not relevant for modeling, and consolidated the results into a structured dataset. We then matched the expression data to case identifiers and standardized the cancer stage labels so the dataset could support both binary and multi-class classification tasks.

The main challenges we faced were dealing with inconsistencies in the raw TCGA folders and making sure the mapping between expression files and clinical data was accurate. Working together allowed us to share the workload and double-check each step, which helped us ensure the dataset was clean, labeled, and ready for feature selection.

4. Describe the tasks to be completed the coming week.

Next week (Week 4) we will begin feature selection to reduce dimensionality and identify the most informative miRNAs for breast cancer stage classification. With Dr. Ali's guidance, the team will experiment with multiple methods in scikit-learn. Each member will focus on a specific approach: Joshua would work on chi-square regularization, Brehon would apply Neighborhood Component Analysis (NCA), Nicholas would begin exploring deep learning-based feature selection, and Anika would focus on L1 norm-based methods (LASSO). As a group, we would compare the results and evaluate the stability of the selected feature sets.

In parallel, we will also work on drafting a product press release. To divide the work, the sections were assigned as follows: Joshua will create the logo, draft contact information, dateline, and headline; Anika will write the sub-headline and introduction; Brehon will prepare the body and boilerplate; and Nicholas will handle the close and attach photos of individual team members. Although the sections are split, the final press release will be a shared responsibility, and we will all review and edit it together before submission.

Previous Report Summary. Date: 8/25/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	2	0	2
Anika Baro Villa	3	0	2
Nicholas Campos	2	0	1
Brehon Parker	3	0	2

Current Report Summary. Date: 9/01/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo			
Anika Baro Villa	2	0	2
Nicholas Campos			
Brehon Parker			

Name of the Member: Anika Baro Villa

Report Date: 9/01/25

Task 2.1: Download data (miRNA + clinical) from GDC Portal (8/26/25)

I downloaded the miRNA expression files and clinical metadata from the GDC Portal to prepare for cleaning and labeling. Following Dr. Ali Ibrahim's guidance, I applied filters for breast cancer, microRNA sequencing, expression quantification, .txt format, open access, and both normal and tumor samples. I also included clinical metadata in JSON format, confirmed the counts (1,209 files across 1,081 cases) matched Dr. Ali's figures, and downloaded the manifest, cart, and data files.

Task 2.2: Write new sections for the revised project proposal (8/30/25)

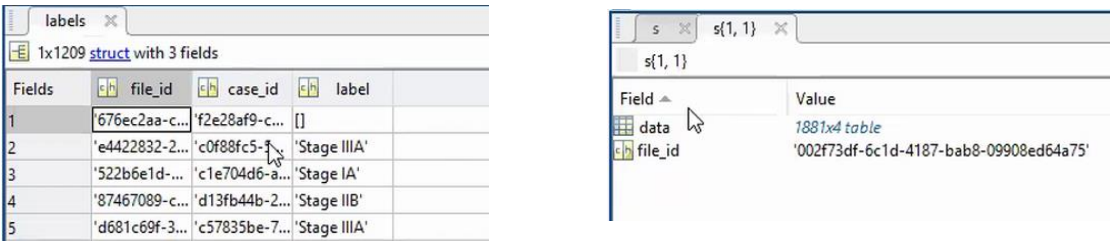
I contributed to the revised project proposal by drafting key sections, including a summary of completed tasks, use-case scenarios with test cases, an updated WBS table reflecting current timelines and responsibilities, and a detailed task breakdown outlining my responsibilities in the project.

Task 2.3: Create shared folders for ED2 deliverables and update project planner (8/30/25)

I collaborated with the team to create organized shared folders in Microsoft Teams for ED2 deliverables and progress reports. Additionally, I added tasks to my assigned bucket in the project planner for the next 14 weeks, aligning them with the updated WBS to support consistent weekly tracking of responsibilities and progress.

Task 3.1: Clean miRNA expression files (9/1/25)

This task focused on cleaning and organizing the raw TCGA miRNA expression files to prepare them for integration with clinical metadata. Using MATLAB scripts developed by Dr. Ali Ibrahim from his previous research, we automated the process to ensure consistency and accuracy. Each folder in the dataset corresponded to a unique file_id and contained a .txt file with multiple columns of expression data, along with occasional annotation files. We extracted the third column, reads_per_million_miRNA_mapped, as the measure of expression because it represents normalized counts appropriate for downstream analysis. We also removed extra annotation files that did not contain expression values, which reduced the dataset from 1,244 to 1,209 usable files. The scripts then generated two main outputs. The first was the labels struct, created by combining information from two cart JSON files. The metadata JSON gave us the link between file_id and case_id, and the clinical JSON provided the ajcc_pathologic_stage, which we matched to each sample using case_id. The second was S, a matrix that stored the expression values extracted from the .txt files together with their corresponding file_ids.



Fields	file_id	case_id	label
1	'676ec2aa-c...	'f2e28af9-c...	{} (empty)
2	'e4422832-2...	'c0f88fc5-f...	'Stage IIIA'
3	'522b6e1d-...	'c1e704d6-a...	'Stage IA'
4	'87467089-c...	'd13fb44b-2...	'Stage IIB'
5	'd681c69f-3...	'c57835be-7...	'Stage IIIA'

Field	Value
data	1881x4 table
file_id	'002f73df-6c1d-4187-bab8-09908ed64a75'

Task 3.2: Link cleaned miRNA expression data with clinical metadata by case IDs (9/1/25)

Using MATLAB scripts developed by Dr. Ali Ibrahim, we matched each file_id from the expression matrix S to its associated case_id in the labels struct, which also contained the disease and staging information extracted from the clinical JSON. We then standardized the stage labels by normalizing the text strings and collapsing all sub-stages into their parent stages (for example, IA and IB were grouped into Stage I, and similarly for the other stages). After harmonization, the stages were encoded into numeric values, with 0 representing normal tissue and 1–4 representing stages I through IV. The final dataset contained 1,209 samples and 1,882 columns, of which 1,881 were quantitative miRNA expression features; the last column was the encoded stage label. When constructing this table, the miRNA identifiers from column 1 of the expression files were applied as feature headers, ensuring consistent ordering across all samples. The merged dataset was then exported to CSV with these headers and the Label column appended at the end as show below.

	A	B	C	D	E	F	G	
1	hsa-let-7a	hsa-let-7a	hsa-let-7a	hsa-let-7b	hsa-let-7c	hsa-let-7c	hsa-let-7e	
2	6743.308	6822.205	6678.756	10892.58	1735.735	936.0061	1534.907	
3	11967.89	11974.72	11990.72	20124.83	2525.844	367.3567	1640.092	
4	12660.72	12466.33	12601.95	15830.67	576.1266	722.7952	2102.586	
5	6664.506	6583.932	6699.674	15931.42	1878.58	551.5545	919.2576	
6	5652.188	5872.823	5890.159	14259.31	1534.199	436.5415	620.9291	
7	8395.1	8287.75	8330.529	20291.59	510.1147	539.1719	1337.437	
8	19889.39	19847.31	19961.06	147288.5	836.3958	596.3923	827.8478	
9	9772.399	9833.226	9719.042	62556.16	428.9931	1037.8	1157.855	
10	30418.36	30303.24	30351.26	47637.62	6336.517	339.2199	1992.224	

	BTF	BTG	BTH	BTI	BTJ
hsa-mir-9	hsa-mir-9	hsa-mir-9	hsa-mir-9	hsa-mir-9	Label
32.27607	136.2768	472.187	44750.18		0
66.5594	47.14624	881.4854	36506.55		3
206.9435	84.887	158.7236	43876.53		1
25.37418	46.74191	473.6514	23033.52		2
24.42741	52.79473	556.3146	43971.71		3
30.67146	78.29292	136.4073	49869.36		0
127.5635	48.65825	253.1544	81071.22		2
22.41009	61.36095	132.3262	110513.7		2
27.08834	46.17331	1731.191	28544.65		2

Task 4.1 Apply feature selection (LASSO regularization) to identify the most informative miRNAs (9/14/25)

In this task, we will apply several feature selection methods to reduce dimensionality and identify the most informative miRNAs for breast cancer stage classification. Under Dr. Ali’s guidance, the team will explore techniques such as univariate tests, mutual information, RFE, and Sequential Selection, but my focus will be on L1 norm–based

methods (LASSO regularization). The goal is to select stable subsets of miRNAs that improve both interpretability and predictive performance.

Task 4.2 Draft a product press release (sub-headline and introduction) (9/14/25)

Our team will work on drafting a product press release that would include all standard sections, such as logo, contact, dateline, headline, sub-headline, introduction, body, boilerplate, close, and team photos. The work was split across team members, and my responsibility would be drafting the sub-headline and introduction, which should capture attention and clearly communicate the purpose of our system.